Log Book Entry 4.2

NB2420 Electronic Instrumentation · Manual 4.2 · Real opamps

Hand in via BrightSpace

Name(s): Partner:

Date: Group:

PREPARATION

complete before the practicum

1. *What theories, laws, and/or formulas are most important for this experiment?*
2. *What do you expect to learn?*
3. *What do you expect to observe, and why?*

BIAS AND OFFSET

Tasks A1, A2, S1, I1, C1

Your own chip, not the typical one. The class spread column is the point: it tells you whether your number is normal.

Quantity	Datasheet typ	Datasheet max	Simulated	Measured (your chip)	Class spread
V_{offset}					
I_{bias}					

Which R_a did you choose to isolate each, and why that value?

BANDWIDTH AND SLEW RATE

Tasks A3, A4, S2, I2, I3, C2

Fill one row per amplifier you designed. If GBW is really constant, the last column should not move.

Closed-loop gain	Predicted BW	Simulated BW	Measured BW	Gain × BW

Slew rate measured:

Slew-limited frequency, predicted and measured:

Output current limit found (Task I3):

GOLDEN RULES, TIGHTENED

Tasks I4, I5, C3

Rewrite each ideal rule from 4.1 as an inequality, using the number you actually measured.

Golden rule (4.1)	**Tightened version (4.2)**
$U_+ = U_-$	
$I_{in+} = I_{in-} = 0$	
The output is an ideal voltage source	
Open-loop gain $\to \infty$	
No rate limit on the output	

Integrator Tasks A6, S3, I4, I5

Drift rate before taming:

What you added to tame it:

Slope before and after scaling:

REFLECTION

complete before submission

1. *Which results matched or differed from your expectations, and why?*
2. *What is the most important thing you learned?*
3. *What would you do differently next time?*

TA check-off: Date: